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Method and system for automated personal training that includes training programs

Abstract

Systems and methods for creating personalized exercise programs are disclosed. An image capture device and a computer device are used to capture images of a user while the user performs athletic movements. The images may then be evaluated to create a human movement screen score. The human movement screen score, goal and time commitment information may then be used to create a personalized exercise program tailored to the specific user.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 15/255,566, filed Sep. 2, 2016, which is a continuation of U.S. patent application Ser. No. 13/304,064, filed Nov. 23, 2011, now U.S. Pat. No. 9,457,256, which is a continuation-in-part of U.S. patent application Ser. No. 13/290,359, filed Nov. 7, 2011, now U.S. Pat. No. 9,283,429, which claims the benefit of, and priority to, U.S. Provisional Patent Application Nos. 61/410,777 filed Nov. 5, 2010, 61/417,102 filed Nov. 24, 2010, 61/422,511 filed Dec. 13, 2010, 61/432,472 filed Jan. 13, 2011, and 61/433,792 filed Jan. 18, 2011, each of which is entitled “Method and System for Automated Personal Training.” The content of each of the applications is expressly incorporated herein by reference in its entirety for any and all non-limiting purposes.

BACKGROUND

(1) While most people appreciate the importance of physical fitness, many have difficulty finding the motivation required to maintain a regular exercise program. Some people find it particularly difficult to maintain an exercise regimen that involves continuously repetitive motions, such as running, walking and bicycling.

(2) Additionally, individuals may view exercise as work or a chore and thus, separate it from enjoyable aspects of their daily lives. Often, this separation between athletic activity and other activities reduces the amount of motivation that an individual might have toward exercising. Further, athletic activity services and systems directed toward encouraging individuals to engage in athletic activities might also be too focused on one or more particular activities while an individual's interests are ignored. This may further decrease a user's interest in participating in athletic activities or using the athletic activity services and systems.

(3) Therefore, improved systems and methods to address these and other shortcomings in the art are desired.

BRIEF SUMMARY

(4) The following presents a simplified summary in order to provide a basic understanding of some aspects of the disclosure. The summary is not an extensive overview of the disclosure. It is neither intended to identify key or critical elements of the disclosure nor to delineate the scope of the disclosure. The following summary merely presents some concepts of the disclosure in a simplified form as a prelude to the description below.

(5) Aspects of the invention provide systems and methods for creating personalized exercise programs. A computer device, such as a video game console may be used with an image capture device, such as a group of cameras to capture images of a user performing athletic movements. As used herein, an “athletic movement” includes movements relating to fitness, exercise, flexibility, including movements that may be part of one or more single and multiple participant athletic competitions, exercise routines, and/or combinations thereof. The images may then be evaluated to create a human movement screen score. The human movement screen score may be used to create a personalized exercise program tailored to the specific user. A human movement screen (HMS) is a ranking and grading system that documents movement patterns that may be key to normal function. The functional movement screen (FMS) developed by Gray Cook is an example of a human movement screen.

(6) In some embodiments the user may also provide preference data, such as data relating to time commitments, preferred exercises and a preferred number of exercise sessions in a predetermined time period. The computer device may consider these factors when creating a personalized exercise program.

(7) Certain other embodiments may capture athletic movement data with accelerometers, gyroscopes or position locating devices, such as GPS devices.

(8) In other embodiments, the present invention can be partially or wholly implemented on a tangible non-transitory computer-readable medium, for example, by storing computer-executable instructions or modules, or by utilizing computer-readable data structures.

(9) Of course, the methods and systems of the above-referenced embodiments may also include other additional elements, steps, computer-executable instructions, or computer-readable data structures.

(10) These and other aspects of the embodiments are discussed in greater detail throughout this disclosure, including the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is illustrated by way of example and not limited in the accompanying figures in which like reference numerals indicate similar elements and in which:

(2) FIGS. 1A-B illustrate an example of a system for providing personal training in accordance with example embodiments, wherein FIG. 1A illustrates an example network configured to monitor athletic activity, and FIG. 1B illustrates an example computing device in accordance with example embodiments.

(3) FIGS. 2A-B illustrate example sensor assemblies that may be worn by a user in accordance with example embodiments.

(4) FIG. 3 illustrates example points on a user's body to monitor in accordance with example embodiments.

(5) FIG. 4 illustrates an example posture assessment in accordance with example embodiments.

(6) FIG. 5 illustrates an exemplary method that may be used to generate personalized training programs in accordance with an embodiment of the invention

(7) FIG. 6 shows exemplary instructions that may be provided to a user to perform the athletic movement.

(8) FIG. 7 illustrates exemplary personalized exercise program phases in accordance with an embodiment of the invention.

(9) FIG. 8 illustrates another example flow diagram for a user's initial interaction to determine a user's baseline physical fitness level.

(10) FIG. 9 illustrates an example flow diagram creating a personalized exercise program

(11) FIG. 10 illustrates an example graphical user interface with options to select a trainer, to start a new program, or to do a drop-in workout session.

(12) FIG. 11 illustrates an example graphical user interface prompting a user to input user desired workout selections.

(13) FIG. 12 illustrates an example graphical user interface that includes a scroll bar permitting a user to select a desired number of times to exercise per week.

(14) FIG. 13 illustrates an example graphical user interface permitting a user to input reminders so that an email or text message may be sent to a device (e.g., mobile phone) to remind the user about an upcoming workout session.

(15) FIG. 14 illustrates an example initial baseline physical fitness level based on human movement screen scoring and tempo.

(16) FIG. 15 illustrates example drills for assessing user performance relative to performance pillars and body movement categories.

(17) FIG. 16 illustrates an example relationship between body movement categories and human movement screen exercises.

(18) FIG. 17 provides an example of different tempos at which a same drill can be performed.

(19) FIG. 18 illustrates an example workout structure.

(20) FIG. 19 illustrates an example flow diagram for setting goals and encouraging commitment from a user.

(21) FIG. 20-34 illustrates an example six month workout plan that may include a baseline workout

and six month long programs.

(22) FIG. 35 illustrates an example 4 week routine to aid a user in achieving a goal.

(23) FIG. 36 illustrates an example leading a user through a workout session.

(24) FIG. 37 illustrates an example graphical user interface prompting a user to begin a workout session, and asking how long the user has to work out.

(25) FIG. 38 illustrates an example graphical user interface leading a user through a warm up session.

(26) FIG. 39 illustrates an example graphical user interface providing a demonstration of a first exercise drill.

(27) FIG. 40 illustrates an example graphical user interface displaying an image of the user performing a drill.

(28) FIG. 41 illustrates an example graphical user interface comparing a user's form versus desired form.

(29) FIG. 42 illustrates an example graphical user interface including an image of a user exercising with added straight lines to show proper back and hip posture during a deep squat.

(30) FIGS. 43A-B illustrate example graphical user interfaces providing a user with feedback on their form and removing the corrective feedback when the user's form improves.

(31) FIG. 44 illustrates an example graphical user interface informing a user of a next training category during the workout session.

(32) FIG. 45 illustrates example data points used by a computer to determine feedback and motivation to provide to a user during a workout session.

(33) FIGS. 46-49 illustrate examples of revising a workout session based on the amount of time a user can commit to a workout.

(34) FIG. 50 illustrates an example flow diagram for providing a user with post-workout information.

(35) FIG. 51 illustrates an example graphical user interface informing a user that a workout session is complete.

(36) FIG. 52-53 illustrates example graphical user interfaces informing a user of their workout performance and an amount of points they have received during the workout.

(37) FIG. 54 illustrates an example graphical user interface prompting a user to continue exercising.

(38) FIG. 55 illustrates an example flow diagram of a drop-in workout session.

(39) FIG. 56 illustrates an example graphical user interface permitting a user to select a drop-in workout session.

(40) FIG. 57 illustrates an example graphical user interface prompting a user to input how long they have to work out during a drop-in session.

(41) FIG. 58 illustrates an example graphical user interface prompting a user to input what type of session they want to do during a drop-in session.

(42) FIG. 59 illustrates an example graphical user interface of an image captured of a user working out during a drop-in session.

(43) FIG. 60 illustrates an exemplary challenge drill.

DETAILED DESCRIPTION

(44) In the following description of the various embodiments, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration various embodiments in which the disclosure may be practiced. It is to be understood that other embodiments may be utilized and structural and functional modifications may be made without departing from the scope and spirit of the present disclosure. Further, headings within this disclosure should not be considered as limiting aspects of the disclosure. Those skilled in the art with the benefit of this disclosure will appreciate that the example embodiments are not limited to the example headings.

I. Example Personal Training System

(45) A. Illustrative Computing Devices

(46) FIG. 1A illustrates an example of a personal training system **100** in accordance with example embodiments. Example system **100** may include one or more electronic devices, such as computer **102**. Computer **102** may comprise a mobile terminal, such as a telephone, music player, tablet, netbook or any portable device. In other embodiments, computer **102** may comprise a set-top box (STB), desktop computer, digital video recorder(s) (DVR), computer server(s), and/or any other desired computing device. In certain configurations, computer **102** may comprise a gaming console, such as for example, a Microsoft® XBOX, Sony® Playstation, and/or a Nintendo® Wii gaming consoles. Those skilled in the art will appreciate that these are merely example consoles for descriptive purposes and this disclosure is not limited to any console or device.

(47) Turning briefly to FIG. 1B, computer **102** may include computing unit **104**, which may comprise at least one processing unit **106**. Processing unit **106** may be any type of processing device for executing software instructions, such as for example, a microprocessor device. Computer **102** may include a variety of non-transitory computer readable media, such as memory **108**. Memory **108** may include, but is not limited to, random access memory (RAM) such as RAM **110**, and/or read only memory (ROM), such as ROM **112**. Memory **108** may include any of: electronically erasable programmable read only memory (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic storage devices, or any other medium that can be used to store the desired information and that can be accessed by computer **102**.

(48) The processing unit **106** and the system memory **108** may be connected, either directly or indirectly, through a bus **114** or alternate communication structure to one or more peripheral devices. For example, the processing unit **106** or the system memory **108** may be directly or indirectly connected to additional memory storage, such as a hard disk drive **116**, a removable magnetic disk drive, an optical disk drive **118**, and a flash memory card, as well as to input devices **120**, and output devices **122**. The processing unit **106** and the system memory **108** also may be directly or indirectly connected to one or more input devices **120** and one or more output devices **122**. The output devices **122** may include, for example, a display device **136**, television, printer, stereo, or speakers. In some embodiments one or more display devices may be incorporated into eyewear. The display devices incorporated into eyewear may provide feedback to users. Eyewear incorporating one or more display devices also provides for a portable display system. The input devices **120** may include, for example, a keyboard, touch screen, a remote control pad, a pointing device (such as a mouse, touchpad, stylus, trackball, or joystick), a scanner, a camera or a microphone. In this regard, input devices **120** may comprise one or more sensors configured to sense, detect, and/or measure athletic movement from a user, such as user **124**, shown in FIG. 1A.

(49) Looking again to FIG. 1A, image-capturing device **126** and/or sensor **128** may be utilized in detecting and/or measuring athletic movements of user **124**. In one embodiment, data obtained from image-capturing device **126** or sensor **128** may directly detect athletic movements, such that the data obtained from image-capturing device **126** or sensor **128** is directly correlated to a motion parameter. For example, and with reference to FIG. 4, image data from image-capturing device **126** may detect that the distance between sensor locations **402g** and **402i** has decreased and therefore, image-capturing device **126** alone may be configured to detect that user's **124** right arm has moved. Yet, in other embodiments, data from image-capturing device **126** and/or sensor **128** may be utilized in combination, either with each other or with other sensors to detect and/or measure movements. Thus, certain measurements may be determined from combining data obtained from two or more devices. Image-capturing device **126** and/or sensor **128** may include or be operatively connected to one or more sensors, including but not limited to: an accelerometer, a gyroscope, a location-determining device (e.g., GPS), light sensor, temperature sensor (including ambient temperature and/or body temperature), heart rate monitor, image-capturing sensor, moisture sensor

and/or combinations thereof. Example uses of illustrative sensors **126**, **128** are provided below in Section I.C, entitled “Illustrative Sensors.” Computer **102** may also use touch screens or image capturing device to determine where a user is pointing to make selections from a graphical user interface. One or more embodiments may utilize one or more wired and/or wireless technologies, alone or in combination, wherein examples of wireless technologies include Bluetooth® technologies, Bluetooth® low energy technologies, and/or ANT technologies.

(50) B. Illustrative Network

(51) Still further, computer **102**, computing unit **104**, and/or any other electronic devices may be directly or indirectly connected to one or more network interfaces, such as example interface **130** (shown in FIG. **1B**) for communicating with a network, such as network **132**. In the example of FIG. **1B**, network interface **130**, may comprise a network adapter or network interface card (NIC) configured to translate data and control signals from the computing unit **104** into network messages according to one or more communication protocols, such as the Transmission Control Protocol (TCP), the Internet Protocol (IP), and the User Datagram Protocol (UDP). These protocols are well known in the art, and thus will not be discussed here in more detail. An interface **130** may employ any suitable connection agent for connecting to a network, including, for example, a wireless transceiver, a power line adapter, a modem, or an Ethernet connection. Network **132**, however, may be any one or more information distribution network(s), of any type(s) or topography(s), alone or in combination(s), such as internet(s), intranet(s), cloud(s), LAN(s). Network **132** may be any one or more of cable, fiber, satellite, telephone, cellular, wireless, etc. Networks are well known in the art, and thus will not be discussed here in more detail. Network **132** may be variously configured such as having one or more wired or wireless communication channels to connect one or more locations (e.g., schools, businesses, homes, consumer dwellings, network resources, etc.), to one or more remote servers **134**, or to other computers, such as similar or identical to computer **102**. Indeed, system **100** may include more than one instance of each component (e.g., more than one computer **102**, more than one display **136**, etc.).

(52) Regardless of whether computer **102** or other electronic device within network **132** is portable or at a fixed location, it should be appreciated that, in addition to the input, output and storage peripheral devices specifically listed above, the computing device may be connected, such as either directly, or through network **132** to a variety of other peripheral devices, including some that may perform input, output and storage functions, or some combination thereof. In certain embodiments, a single device may integrate one or more components shown in FIG. **1A**. For example, a single device may include computer **102**, image-capturing device **126**, sensor **128**, display **136** and/or additional components. In one embodiment, sensor device **138** may comprise a mobile terminal having a display **136**, image-capturing device **126**, and one or more sensors **128**. Yet, in another embodiment, image-capturing device **126**, and/or sensor **128** may be peripherals configured to be operatively connected to a media device, including for example, a gaming or media system. Thus, it goes from the foregoing that this disclosure is not limited to stationary systems and methods. Rather, certain embodiments may be carried out by a user **124** in almost any location.

(53) C. Illustrative Sensors

(54) Computer **102** and/or other devices may comprise one or more sensors **126**, **128** configured to detect and/or monitor at least one fitness parameter of a user **124**. Sensors **126** and/or **128**, may include but not limited to: an accelerometer, a gyroscope, a location-determining device (e.g., GPS), light sensor, temperature sensor (including ambient temperature and/or body temperature), sleep pattern sensors, heart rate monitor, image-capturing sensor, moisture sensor and/or combinations thereof. Network **132** and/or computer **102** may be in communication with one or more electronic devices of system **100**, including for example, display **136**, an image capturing device **126** (e.g., one or more video cameras), and sensor **128**, which may be an infrared (IR) device. In one embodiment sensor **128** may comprise an IR transceiver. For example, sensors **126**, and/or **128** may transmit waveforms into the environment, including towards the direction of user

124 and receive a “reflection” or otherwise detect alterations of those released waveforms. In yet another embodiment, image-capturing device **126** and/or sensor **128** may be configured to transmit and/or receive other wireless signals, such as radar, sonar, and/or audible information. Those skilled in the art will readily appreciate that signals corresponding to a multitude of different data spectrums may be utilized in accordance with various embodiments. In this regard, sensors **126** and/or **128** may detect waveforms emitted from external sources (e.g., not system **100**). For example, sensors **126** and/or **128** may detect heat being emitted from user **124** and/or the surrounding environment. Thus, image-capturing device **126** and/or sensor **128** may comprise one or more thermal imaging devices. In one embodiment, image-capturing device **126** and/or sensor **128** may comprise an IR device configured to perform range phenomenology. As a non-limited example, image-capturing devices configured to perform range phenomenology are commercially available from Flir Systems, Inc. of Portland, Oregon. Although image capturing device **126** and sensor **128** and display **136** are shown in direct (wirelessly or wired) communication with computer **102**, those skilled in the art will appreciate that any may directly communicate (wirelessly or wired) with network **132**.

(55) 1. Multi-Purpose Electronic Devices

(56) User **124** may possess, carry, and/or wear any number of electronic devices, including sensory devices **138**, **140**, **142**, and/or **144**. In certain embodiments, one or more devices **138**, **140**, **142**, **144** may not be specially manufactured for fitness or athletic purposes. Indeed, aspects of this disclosure relate to utilizing data from a plurality of devices, some of which are not fitness devices, to collect, detect, and/or measure athletic data. In one embodiment, device **138** may comprise a portable electronic device, such as a telephone or digital music player, including an IPOD®, IPAD®, or iPhone®, brand devices available from Apple, Inc. of Cupertino, California or Zune® or Microsoft® Windows devices available from Microsoft of Redmond, Washington. As known in the art, digital media players can serve as both an output device for a computer (e.g., outputting music from a sound file or pictures from an image file) and a storage device. In one embodiment, device **138** may be computer **102**, yet in other embodiments, computer **102** may be entirely distinct from device **138**. Regardless of whether device **138** is configured to provide certain output, it may serve as an input device for receiving sensory information. Devices **138**, **140**, **142**, and/or **144** may include one or more sensors, including but not limited to: an accelerometer, a gyroscope, a location-determining device (e.g., GPS), light sensor, temperature sensor (including ambient temperature and/or body temperature), heart rate monitor, image-capturing sensor, moisture sensor and/or combinations thereof. In certain embodiments, sensors may be passive, such as reflective materials that may be detected by image-capturing device **126** and/or sensor **128** (among others). In certain embodiments, sensors **144** may be integrated into apparel, such as athletic clothing. For instance, the user **124** may wear one or more on-body sensors **144a-b**. Sensors **144** may be incorporated into the clothing of user **124** and/or placed at any desired location of the body of user **124**. Sensors **144** may communicate (e.g., wirelessly) with computer **102**, sensors **128**, **138**, **140**, and **142**, and/or camera **126**. Examples of interactive gaming apparel are described in U.S. patent application Ser. No. 10/286,396, filed Oct. 30, 2002, and published as U.S. Pat. Pub. No. 2004/0087366, the contents of which are incorporated herein by reference in its entirety for any and all non-limiting purposes. In certain embodiments, passive sensing surfaces may reflect waveforms, such as infrared light, emitted by image-capturing device **126** and/or sensor **128**. In one embodiment, passive sensors located on user's **124** apparel may comprise generally spherical structures made of glass or other transparent or translucent surfaces which may reflect waveforms. Different classes of apparel may be utilized in which a given class of apparel has specific sensors configured to be located proximate to a specific portion of the user's **124** body when properly worn. For example, golf apparel may include one or more sensors positioned on the apparel in a first configuration and yet soccer apparel may include one or more sensors positioned on apparel in a second configuration.

(57) Devices **138-144** may communicate with each other, either directly or through a network, such as network **132**. Communication between one or more of devices **138-144** may communicate through computer **102**. For example, two or more of devices **138-144** may be peripherals operatively connected to bus **114** of computer **102**. In yet another embodiment, a first device, such as device **138** may communicate with a first computer, such as computer **102** as well as another device, such as device **142**, however, device **142** may not be configured to connect to computer **102** but may communicate with device **138**. Those skilled in the art will appreciate that other configurations are possible.

(58) Some implementations of the example embodiments may alternately or additionally employ computing devices that are intended to be capable of a wide variety of functions, such as a desktop or laptop personal computer. These computing devices may have any combination of peripheral devices or additional components as desired. Also, the components shown in FIG. **1B** may be included in the server **134**, other computers, apparatuses, etc.

(59) 2. Illustrative Apparel/Accessory Sensors

(60) In certain embodiments, sensory devices **138**, **140**, **142** and/or **144** may be formed within or otherwise associated with user's **124** clothing or accessories, including a watch, armband, wristband, necklace, shirt, shoe, or the like. Examples of shoe-mounted and wrist-worn devices (devices **140** and **142**, respectively) are described immediately below, however, these are merely example embodiments and this disclosure should not be limited to such.

(61) i. Shoe-Mounted Device

(62) In certain embodiments, sensory device **140** may comprise footwear which may include one or more sensors, including but not limited to: an accelerometer, location-sensing components, such as GPS, and/or a force sensor system. FIG. **2A** illustrates one exemplary embodiment of an example sensor system **202**. In certain embodiments, system **202** may include a sensor assembly **204**. Assembly **204** may comprise one or more sensors, such as for example, an accelerometer, location-determining components, and/or force sensors. In the illustrated embodiment, assembly **204** incorporates a plurality of sensors, which may include force-sensitive resistor (FSR) sensors **206**. In yet other embodiments, other sensor(s) may be utilized. Port **208** may be positioned within a sole structure **209** of a shoe. Port **208** may optionally be provided to be in communication with an electronic module **210** (which may be in a housing **211**) and a plurality of leads **212** connecting the FSR sensors **206** to the port **208**. Module **210** may be contained within a well or cavity in a sole structure of a shoe. The port **208** and the module **210** include complementary interfaces **214**, **216** for connection and communication.

(63) In certain embodiments, at least one force-sensitive resistor **206** shown in FIG. **2A** may contain first and second electrodes or electrical contacts **218**, **220** and a force-sensitive resistive material **222** and/or **224** disposed between the electrodes **218**, **220** to electrically connect the electrodes **218**, **220** together. When pressure is applied to the force-sensitive material **222/224**, the resistivity and/or conductivity of the force-sensitive material **222/224** changes, which changes the electrical potential between the electrodes **218**, **220**. The change in resistance can be detected by the sensor system **202** to detect the force applied on the sensor **216**. The force-sensitive resistive material **222/224** may change its resistance under pressure in a variety of ways. For example, the force-sensitive material **222/224** may have an internal resistance that decreases when the material is compressed, similar to the quantum tunneling composites described in greater detail below. Further compression of this material may further decrease the resistance, allowing quantitative measurements, as well as binary (on/off) measurements. In some circumstances, this type of force-sensitive resistive behavior may be described as “volume-based resistance,” and materials exhibiting this behavior may be referred to as “smart materials.” As another example, the material **222/224** may change the resistance by changing the degree of surface-to-surface contact. This can be achieved in several ways, such as by using microprojections on the surface that raise the surface resistance in an uncompressed condition, where the surface resistance decreases when the

microprojections are compressed, or by using a flexible electrode that can be deformed to create increased surface-to-surface contact with another electrode. This surface resistance may be the resistance between the material **222** and the electrode **218**, **220** and/or the surface resistance between a conducting layer (e.g. carbon/graphite) and a force-sensitive layer (e.g. a semiconductor) of a multi-layer material **222/224**. The greater the compression, the greater the surface-to-surface contact, resulting in lower resistance and enabling quantitative measurement. In some circumstances, this type of force-sensitive resistive behavior may be described as “contact-based resistance.” It is understood that the force-sensitive resistive material **222/224**, as defined herein, may be or include a doped or non-doped semiconducting material.

(64) The electrodes **218**, **220** of the FSR sensor **206** can be formed of any conductive material, including metals, carbon/graphite fibers or composites, other conductive composites, conductive polymers or polymers containing a conductive material, conductive ceramics, doped semiconductors, or any other conductive material. The leads **212** can be connected to the electrodes **218**, **220** by any suitable method, including welding, soldering, brazing, adhesively joining, fasteners, or any other integral or non-integral joining method. Alternately, the electrode **218**, **220** and associated lead(s) **212** may be formed of a single piece of the same material **222/224**. In further embodiments, material **222** is configured to have at least one electric property (e.g., conductivity, resistance, etc.) than material **224**. Examples of exemplary sensors are disclosed in U.S. patent application Ser. No. 12/483,824, filed on Jun. 12, 2009, the contents of which are incorporated herein in their entirety for any and all non-limiting purposes.

(65) ii. Wrist-Worn Device

(66) As shown in FIG. 2B, device **226** (which may be, or be a duplicative of or resemble sensory device **142** shown in FIG. 1A) may be configured to be worn by user **124**, such as around a wrist, arm, ankle or the like. Device **226** may monitor movements of a user, including, e.g., athletic movements or other activity of user **124**. For example, in one embodiment, device **226** may be activity monitor that measures, monitors, tracks or otherwise senses the user's activity (or inactivity) regardless of the user's proximity or interactions with computer **102**. Device **226** may detect athletic movement or other activity (or inactivity) during user's **124** interactions with computer **102** and/or operate independently of computer **102**. Device **226** may communicate directly or indirectly, wired or wirelessly, with network **132** and/or other devices, such as devices **138** and/or **140**. Athletic data obtained from device **226** may be utilized in determinations conducted by computer **102**, such as determinations relating to which exercise programs are presented to user **124**. As used herein, athletic data means data regarding or relating to a user's activity (or inactivity). In one embodiment, device **226** may wirelessly interact with a remote website such as a site dedicated to fitness or health related subject matter, either directly or indirectly (e.g., via a mobile device, such as device **138** associated with user **124**). In this or another embodiment, device **226** may interact with a mobile device, such as device **138**, as to an application dedicated to fitness or health related subject matter. In these or other embodiments, device **226** may interact with both a mobile device as to an application as above, such as device **138**, and a remote website, such as a site dedicated to fitness or health related subject matter, either directly or indirectly (e.g., via the mobile device, such as device **138**). In some embodiments, at some predetermined time(s), the user may wish to transfer data from the device **226** to another location. For example, a user may wish to upload data from a portable device with a relatively smaller memory to a larger device with a larger quantity of memory. Communication between device **226** and other devices may be done wirelessly and/or through wired mechanisms.

(67) As shown in FIG. 2B, device **226** may include an input mechanism, such as a button **228**, to assist in operation of the device **226**. The button **228** may be a depressible input operably connected to a controller **230** and/or any other electronic components, such as one or more elements of the type(s) discussed in relation to computer **102** shown in FIG. 1B. Controller **230** may be embedded or otherwise part of housing **232**. Housing **232** may be formed of one or more

materials, including elastomeric components and comprise one or more displays, such as display **234**. The display may be considered an illuminable portion of the device **226**. The display **234** may include a series of individual lighting elements or light members such as LED lights **234** in an exemplary embodiment. The LED lights may be formed in an array and operably connected to the controller **230**. Device **226** may include an indicator system **236**, which may also be considered a portion or component of the overall display **234**. It is understood that the indicator system **236** can operate and illuminate in conjunction with the display **234** (which may have pixel member **235**) or completely separate from the display **234**. The indicator system **236** may also include a plurality of additional lighting elements or light members **238**, which may also take the form of LED lights in an exemplary embodiment. In certain embodiments, indicator system **236** may provide a visual indication of goals, such as by illuminating a portion of lighting members **238** to represent accomplishment towards one or more goals.

(68) A fastening mechanism **240** can be unlatched wherein the device **226** can be positioned around a wrist of the user **124** and the fastening mechanism **240** can be subsequently placed in a latched position. The user can wear the device **226** at all times if desired. In one embodiment, fastening mechanism **240** may comprise an interface, including but not limited to a USB port, for operative interaction with computer **102** and/or devices **138**, **140**, and/or recharging an internal power source.

(69) In certain embodiments, device **226** may comprise a sensor assembly (not shown in FIG. 2B). The sensor assembly may comprise a plurality of different sensors. In an example embodiment, the sensor assembly may comprise or permit operative connection to an accelerometer (including in the form of a multi-axis accelerometer), a gyroscope, a location-determining device (e.g., GPS), light sensor, temperature sensor (including ambient temperature and/or body temperature), heart rate monitor, image-capturing sensor, moisture sensor and/or combinations thereof. Detected movements or parameters from device's **142** sensor(s), may include (or be used to form) a variety of different parameters, metrics or physiological characteristics including but not limited to speed, distance, steps taken, and energy expenditure such as calories, heart rate and sweat detection. Such parameters may also be expressed in terms of activity points or currency earned by the user based on the activity of the user. Examples of wrist-worn sensors that may be utilized in accordance with various embodiments are disclosed in U.S. patent application Ser. No. 13/287,064, filed on Nov. 1, 2011, the contents of which are incorporated herein in their entirety for any and all non-limiting purposes.

(70) iii. Identify Sensory Locations

(71) The system **100** may process sensory data to identify user movement data. In one embodiment, sensory locations may be identified. For example, images of recorded video, such as from image-capturing device **126**, may be utilized in an identification of user movement. For example, the user may stand a certain distance, which may or may not be predefined, from the image-capturing device **126**, and computer **102** may process the images to identify the user **124** within the video, for example, using disparity mapping techniques. In an example, the image capturing device **126** may be a stereo camera having two or more lenses that are spatially offset from one another and that simultaneously capture two or more images of the user. Computer **102** may process the two or more images taken at a same time instant to generate a disparity map for determining a location of certain parts of the user's body in each image (or at least some of the images) in the video using a coordinate system (e.g., Cartesian coordinates). The disparity map may indicate a difference between an image taken by each of the offset lenses.

(72) In a second example, one or more sensors may be located on or proximate to the user's **124** body at various locations or wear a suit having sensors situated at various locations. Yet, in other embodiments, sensor locations may be determined from other sensory devices, such as devices **138**, **140**, **142** and/or **144**. With reference to FIG. 3, sensors may be placed (or associated with, such as with image-capturing device **126**) body movement regions, such as joints (e.g., ankles, elbows, shoulders, etc.) or at other locations of interest on the user's **124** body. Example sensory

locations are denoted in FIG. 3 by locations **302a-302o**. In this regard, sensors may be physical sensors located on/in a user's clothing, yet in other embodiments, sensor locations **302a-302o** may be based upon identification of relationships between two moving body parts. For example, sensor location **302a** may be determined by identifying motions of user **124** with an image-capturing device, such as image-capturing device **126**. Thus, in certain embodiments, a sensor may not physically be located at a specific location (such as sensor locations **302a-302o**), but is configured to sense properties of that location, such as with image-capturing device **126**. In this regard, the overall shape or portion of a user's body may permit identification of certain body parts. Regardless of whether an image-capturing device, such as camera **126**, is utilized and/or a physical sensor located on the user **124**, such as sensors within or separate from one or more of device(s) **138, 140, 142, 144** are utilized, the sensors may sense a current location of a body part and/or track movement of the body part. In one embodiment, location **302m** may be utilized in a determination of the user's center of gravity (a.k.a, center of mass). For example, relationships between location **302a** and location(s) **302f/302l** with respect to one or more of location(s) **302m-302o** may be utilized to determine if a user's center of gravity has been elevated along the vertical axis (such as during a jump) or if a user is attempting to “fake” a jump by bending and flexing their knees. In one embodiment, sensor location **302n** may be located at about the sternum of user **124**. Likewise, sensor location **302o** may be located approximate to the naval of user **124**. In certain embodiments, data from sensor locations **302m-302o** may be utilized (alone or in combination with other data) to determine the center of gravity for user **124**. In further embodiments, relationships between multiple several sensor locations, such as sensors **302m-302o**, may be utilized in determining orientation of the user **124** and/or rotational forces, such as twisting of user's **124** torso. Further, one or more locations, such as location(s), may be utilized to as a center of moment location. For example, in one embodiment, one or more of location(s) **302m-302o** may serve as a point for a center of moment location of user **124**. In another embodiment, one or more locations may serve as a center of moment of specific body parts or regions.

(73) In certain embodiments, a time stamp to the data collected indicating a specific time when a body part was at a certain location. Sensor data may be received at computer **102** (or other device) via wireless or wired transmission. A computer, such as computer **102** and/or devices **138, 140, 142, 144** may process the time stamps to determine the locations of the body parts using a coordinate system (e.g., Cartesian coordinates) within each (or at least some) of the images in the video. Data received from image-capturing device **126** may be corrected, modified, and/or combined with data received from one or more other devices **138, 140, 142** and **144**.

(74) In a third example, computer **102** may use infrared pattern recognition to detect user movement and locations of body parts of the user **124**. For example, the sensor **128** may include an infrared transceiver, which may be part of image-capturing device **126**, or another device, that may emit an infrared signal to illuminate the user's **124** body using infrared signals. The infrared transceiver **128** may capture a reflection of the infrared signal from the body of user **124**. Based on the reflection, computer **102** may identify a location of certain parts of the user's body using a coordinate system (e.g., Cartesian coordinates) at particular instances in time. Which and how body parts are identified may be predetermined based on a type of exercise a user is requested to perform.

(75) As part of a workout routine, computer **102** may make an initial postural assessment of the user **124** as part of the initial user assessment. With reference to FIG. 4, computer **102** may analyze front and side images of a user **124** to determine a location of one or more of a user's shoulders, upper back, lower back, hips, knees, and ankles. On-body sensors and/or infrared techniques may also be used, either alone or in conjunction with image-capturing device **126**, to determine the locations of various body parts for the postural assessment. For example, computer **102** may determine assessment lines **124a-g** to determine the locations of a various points on a user's body, such as, for example, ankles, knees, hips, upper back, lower back, and shoulders.

(76) 3. Identify Sensory Regions

(77) In further embodiments, system **100** may identify sensor regions. In one embodiment, assessments lines **144a-g** may be utilized to divide the user's body into regions. For example, lines **144b-f** may be horizontal axes. For example, a “shoulders” region **402** may correlate to a body portion having a lower boundary around the user's shoulders (see line **144b**), region **404** may correlate to the body portion between the shoulders (line **144b**) and about half the distance to the hips (see line **144c**) and thus be an “upper back” region, and region **406** may span the area between line **144c** to the hips (see line **144d**) to comprise a “lower back region.” Similarly, region **408** may span the area between the “hips” (line **144d**) and the “knees” (see line **144e**), region **410** may span between lines **144e** and **144f** and region **412** (see “ankles”) may have an upper boundary around line **144f**. Regions **402-412** may be further divided, such as into quadrants, such as by using axes **144a** and **144g**

(78) 4. Categorize Locations or Regions

(79) Regardless of whether specific points (e.g., locations shown in FIG. 3) and/or regions (e.g., regions shown in FIG. 4), body parts or regions that are not proximate to each other may nonetheless be categorized into the same movement category (see, e.g. block **302c**). For example, as shown in FIG. 4, the “upper back”, “hips”, and “ankles” regions **404**, **408**, **412** may be categorized as belonging to a “mobility” category. In another embodiment, the “lower back” and “knees” regions **406**, **410** may be categorized as belonging to a “stability” category. The categorizations are merely examples, and in other embodiments, a location or region may belong to multiple categories. For example, a “center of gravity” region may be formed from regions **404** and **406**. In one embodiment, a “center of gravity” may comprise portions of regions **404** and **406**. In another embodiment, a “center of moment” category may be provided, either independently, or alternatively, as comprising a portion of at least another category. In one embodiment, a single location may be weighted in two or more categories, such as being 10% weighted in a “stability” category and 90% weighted in a “mobility” category.

(80) Computer **102** may also process the image to determine a color of clothing of the user or other distinguishing features to differentiate the user from their surroundings. After processing, computer **102** may identify a location of multiple points on the user's body and track locations of those points, such as locations **302** in FIG. 3. Computer **102** may also prompt the user to answer questions to supplement the postural assessment, such as, for example, age, weight, etc.

II. Creation of Personal Training Programs

A. Overview

(81) FIG. 5 illustrates an exemplary method that may be used to generate personalized training programs in accordance with an embodiment of the invention. First, in step **502**, instructions to perform athletic movements are provided. The instructions may be generated at a video game console and displayed on a display device, such as a television. Exemplary instructions to perform athletic movements are described in detail below. The athletic movements may be used to generate a human movement screen score. The movements may each map to a specific body movement. In one embodiment, the athletic movements include deep squat, hurdle step, in-line lunge, shoulder mobility, active leg raise, push up and rotary stability. FIG. 6 show exemplary instructions that may be provided to a user to perform the athletic movements.

(82) Next, an image capture device may be used to capture images of an athlete performing the athletic movements in step **504**. The image capture device may include multiple cameras. In one embodiment the image capture device includes three cameras and is used to capture movement in three dimensions. Various embodiments may include cameras that capture light in the visible and/or infrared spectrums.

(83) In step **506** it is determined if data from one or more other sensors is available. Other sensors may include an accelerometer worn on the wrist or embedded in or attached to footwear, a gyroscope, a heart rate monitor, a compass, a location tracking device, such as a GPS device,

pressure sensors inserted into footwear or any of the sensors described above that can be used to capture athletic movements and/or athletic performance. The data received from the image capture device and one or more sensors may be used to generate a human movement screen score. When only data from the image capture device is available, in step **508** a human movement screen score is generated with data from the image capture device. When additional sensor data is available, in step **510** a human movement screen score is generated with data from the image capture device and data from more or more additional sensors. In alternative embodiments a human movement screen score may be generated with only data from the image capture device even when other sensor data is available. For example, sensor data may be available but determined not to be credible or below a threshold. In some embodiments the system may also selectively use data from any of the available sensors.

(84) After a human movement screen score is generated, in step **512** a personalized exercise program is generated based on a human movement screen score. The personalized exercise program may be generated via a device, such as a video game console, a server, or computer **102**, that includes one or more processors. The human movement screen score may reveal areas that can be improved and the personalized exercise program may address those areas. FIG. 7 illustrates exemplary personalized exercise program phases in accordance with an embodiment of the invention. Column **702** lists exercises that may be used to generate human movement screen scores for various body movements. Columns **704** and **706** show exemplary criteria that may be used to score each exercise. Two levels are shown for illustration purposes only. Various embodiments may use three, four or more scoring levels. In one embodiment, four scoring levels are used and the levels include (1) experienced pain during the exercise; (2) exercise was not functionally performed; (3) exercise performed acceptably; and (4) exercise performed well. Columns **708a-708c** shown exemplary exercises that may be part of a personalized training program. A personalized exercise program may include exercises that start at different phases based on the relevant human movement screen score. For example, core stability may start at phase 1 and twist may start at phase 2.

(85) In alternative embodiments a user may also provide preference data that is used to generate the personalized exercise program. The preference data may include time commitments, numbers of exercise sessions, preferred days to exercise, preferred exercises and goals. In one embodiment a user may provide access to an electronic calendar, such as one stored on a website, that shows the user's availability to exercise and the personal training system scans the calendar to determine availability and time commitments. The personal training system may look at historical calendar data to determine probable best times and available time commitments or future calendar data to determine actual availability. The personal training system may also be configured to update the exercise program based on the user's actual availability. For example, a user may have an exercise session scheduled for Monday evening and a scan of the user's calendar reveals that the user has an appointment Monday evening that makes exercising not practical. The personal training system may modify the exercise program to reschedule the exercise to another day. Other changes to the exercise program may also be made to keep the user on track to reach goals. The personal training system may even add calendar events to the user's calendar.

(86) Users may exercise at locations away from the personal training system. Exercise data may be captured by a variety of sensors, such as accelerometers worn on the wrist or other body parts. Accelerometers may also be embedded in or attached to footwear or articles of clothing. Other sensors that may be used to capture exercise data away from the personal training system include gyroscopes, location tracking devices, such as a GPS device, heart rate monitors, pressure sensor systems placed in footwear and any of the sensors described above. The captured exercise data may be provided to the personal training system via a network connection or hardware port, such as a USB port. Returning to FIG. 5, in step **514** it is determined whether exercise data has been captured by a sensor while the user was exercising away from the personal training system. Step **514** may

include determining that GPS data that was captured with a mobile phone while a user ran is available. If no sensor data is available, the process ends in step **516**. One skilled in the art will appreciate that the method shown in FIG. **5** is merely exemplary and may be modified to include other steps and various loops. For example, instead of ending in step **516**, the process may wait a predetermined time and repeat step **514**.

(87) When sensor data is received, in step **518**, the personal training system may modify the personalized exercise program based on the exercise data captured by the sensor. Modifications may include one or more changes to the types of exercises or durations of exercises. For example, if the sensor data indicates that the user recently ran, the next session of the personalized exercise program may be modified to not exercise the primary muscle groups involved in running. Other exemplary modifications include reducing the duration or eliminating an exercise session.

B. Illustrative Embodiments

(88) When a user begins an exercise program, the computer **102** may prompt the user to perform a series of exercises in front of an image capturing device. The computer **102** may process the images and assign a score indicating how well the user was able to complete each of the exercises to establish a baseline physical fitness level. When performing an exercise, the computer **102** may instruct the user to position him or herself at a certain distance and orientation relative to an image capturing device. The computer **102** may process each image to identify different parts of the user's body, such as, for example, their head, shoulders, arms, elbows, hands, wrists, torso, hips, knees, ankles, feet, or other body parts. The computer **102** may generate a set of data identifying a location of various body parts within the image. The computer **102** may process the data set to determine a relationship between certain body parts. These relationships may include an angle of one body part relative to another. For example, when the user is doing a squat, the computer **102** may compare the angle of a user's torso with an angle of the user's thigh. In another example, the computer **102** may compare a location of a user's shoulder relative to their elbow and hand during a push up.

(89) The computer **102** may compare the data set to a desired data set for each exercise to monitor the user's form while performing an exercise. The desired data set may include multiple comparison points throughout an exercise. For example, a push up may be divided into four events: (1) the lowest point where the user's chest is nearest to the ground and their arms are bent; (2) a highest point where the user's chest is farthest from the ground and their arms are straightened; (3) an upward event where the user transitions from the lowest point to the highest point; and (4) a downward event where the user transitions from the highest point to the lowest point. The desired data set may specify comparison points for each of these events focusing on certain body parts. For example, at each comparison point during a pushup, the computer **102** may monitor the spacing of the user's hands, the straightness of the user's back, a location of the user's head relative to their torso, the spacing of the user's feet relative to one another, or other aspects. The desired data set may specify desired locations for each body part being monitored during comparison points within an exercise, as well as permitted variations from the desired locations. If the user's body part varies beyond what is permitted, the computer **102** may provide the user with feedback identifying the body part and a correction to the user's form (e.g., back is arched, and not straight, during a pushup).

(90) The computer **102** may also score the user's performance of an exercise. Scoring may be based on the user's form, how quickly the user was able to complete the exercise (e.g., 20 pushups in 60 seconds), a number of repetitions the user completed, the amount of weight the user used during an exercise, or other exercise metrics. In addition to processing the images, the computer **102** may receive data from other sources. For example, the user may run a predetermined distance as measured by a sensor attached to the user (e.g., sensor in a shoe) or global positioning system (GPS) device and may upload the data to the computer **102**. Based on the images and/or data acquired by other sensors, the computer **102** may determine areas of weakness for the user (e.g., inability to do a pull up) and design a workout to help the user improve their overall fitness level.

Score may be a function of a particular drill and may be focused on position, accuracy and correct execution. Scoring may also be based on time and/or a number sets or repetitions within a set time period.

(91) FIG. **8** illustrates an example flow diagram for a user's initial interaction to determine a user's baseline physical fitness level. In this example, the system may scan the user and prompt the user to sign up and purchase automated training. The system **300** may obtain information from the user about weight, gender, and age, and create an initial baseline physical fitness level. The system may lead the user through an initial set of drills, and set a participation goal.

(92) After completing the baseline physical fitness level for the user, the computer **102** may then create an initial personalized program. The initial personalized program may be a function of user input, static assessment of the user, and a human movement screen. User input may include a user's time commitment, as well as number of exercise sessions per week and one or more goals. The status assessment may provide the user with information and coaching on exercises. The human movement screen score may be assessments of the user's performance of the exercise drills.

(93) FIG. **9** illustrates an example of creating a personalized exercise program. The personalized program may be a function of human movement screen scores, the number of exercise sessions per week the user desires, and the goal of the user (e.g., total body fitness, run a marathon, lose weight, etc.). Goals may also include “get strong”, “get lean” and/or “get toned.” Other factors that may be considered include a user's fitness profile that may consider a current assessment of a user's fitness level as well as user preferences. The user may also specify multiple goals, including to be fitter, stronger, faster, centered, etc. The user may specify a fitness level, such as, for example, beginner, intermediate, advanced. The computer **102** may evaluate the user's fitness level over time to confirm the user input fitness level or to adjust exercises based on measured performance rather than the user specified fitness level. The user may also specify a desired length of their program, such as 1 week, 2 weeks, 4 weeks or a custom length.

(94) To obtain these inputs, the computer **102** may present a graphical user interface (GUI) on the display **302** prompting the user to start a new program and to provide input for the initial personalized program, as shown in FIGS. **10-13**. In FIG. **10**, the graphical user interface may present the user with options to select a trainer (e.g., tabs for Josh or Lisa), to start a new program, or to do a drop-in session. In response to selecting to start a new program, the graphical user interface may prompt the user to input user desired workout selections, as depicted in FIG. **11**. The user may, for example, select a program length and a number of sessions per week. The GUI may then present the user with a total number of workout sessions. The GUI may also present the user with options to buddy up or to join a group where a user may be networked (e.g., via the Internet) with at least one other user to exercise at the same time. When a drop in session is selected, drills for a single session may be selected in a manner that appears random to users, i.e. pseudo random. The selection may be based on one or more goals of the user, such as strength and cardio goals. The selection may also be based on time or performance of the user. The system may select drills for as long as the user wishes to exercise.

(95) FIG. **12** shows a scroll bar in the GUI where a user may select a desired number of times to exercise per week. In FIG. **13**, the GUI permits the user to input reminders so that an email or text message may be sent to a device (e.g., mobile phone) to remind the user about an upcoming workout session. The reminder may include a time and date of one or more upcoming workout sessions.

(96) FIG. **14** illustrates an example baseline physical fitness level assessment determined using human movement screen scoring and tempo. The computer **102** may prompt the user to perform one or more exercise drills, and determine a score each of the assessment exercises using human movement screen scoring as well as the tempo of the user completing the exercise drills. The human movement screen scoring may range from 0 to 3, and the tempo may have categories of slow, moderate, normal, and explosive.

(97) The drills may be used for assessing user performance relative to performance pillars and body movement categories, as depicted in FIG. 15. The performance pillars may be assessments designed to analyze a user's strength, power, speed & agility, energy systems, mobility, and regeneration. Body movement drill categories may include assessments of core stability, twist, squat, pull, single-leg balance, push, bend, and lunge.

(98) FIG. 16 illustrates an example relationship between body movement categories and human movement screen exercises. Human movement screen exercises may include rotary stability, overhead squat, shoulder stability, hurdle step, push up, active leg raise, and in-line lunge. Rotary stability may correspond to the core stability and twist body movement categories, overhead squat may correspond to the squat body movement category, shoulder stability may correspond to the pull body movement category, hurdle step may correspond to the single-leg balance body movement category, push up may correspond to the push body movement category, active leg raise may correspond to the bend body movement category, and in-line lunge may correspond to the lunge body movement category.

(99) The computer 102 may instruct the user to perform the same drill at various tempos, as described in FIG. 17. The tempo of the drill can affect which performance pillar is being assessed. In a squat, for example, the computer 102 may assess a user's strength in a slow tempo and a user's power in an explosive tempo. FIG. 18 illustrates four different tempo categories including slow, moderate, normal, and explosive. Slow may involve the user moving down in an exercise over two seconds, holding for one second, and moving up in the next two seconds.

(100) Based on the human movement screen scoring, the computer 102 may generate a workout structure for the user, an example of which is depicted in FIG. 18. The computer 102 may consider the number of exercise sessions the user is willing to do per week in combination with the human movement screen score to generate a workout program. The workouts may focus on one of three activities: an A strength workout, a B cardio and metabolic workout, and a C regeneration workout. If a user is only willing to workout once per week, the computer 102 may assign a strength workout (i.e., $1 \times A$). If the user is willing to workout more than once per week, the computer 102 may consider the human movement screen score for the user. Generally, the computer 102 may differentiate between scores 14 and above and scores 13 and below. For human movement screen scores 14 and above, the computer 102 may assign one strength workout, one cardio and metabolic workout per week, and no regeneration workouts. For human movement screen scores 13 and below, the computer 102 may assign two strength workouts, no cardio and metabolic workout per week, and no regeneration workouts. The computer 102 may also structure workouts for users desiring to workout 3 or more times per week, as shown in FIG. 19. Alternative embodiments of the invention may include other workout structures that include different types of workouts and numbers of sessions.

(101) FIG. 19 illustrates an example flow diagram for setting goals and encouraging commitment from the user. The computer 102 may present the user with a result of the assessment and comments, and prompt the user to select 1 or more goals (e.g., lose weight, gain strength, etc.). Based on the user's selection and the baseline physical fitness level, the computer 102 may recommend a workout plan. The computer 102 may also permit the user to record a statement of their goal. Goals may be tailored to performing certain events and may include training sessions developed for or by professional athletes. For example, the computer 102 may load a training session pack developed for or by a pro athlete (e.g., 'Paula Radcliffe's marathon training') for real life events.

(102) FIG. 20 illustrates an example workout plan. In this example, the workout plan is for a six month exercise program designed to improve the user's human movement screen score over time. Other workout plans could have different durations, such as one month. In some embodiments users are not presented with an entire workout plan when it is created. The leftmost column may indicate the body movements that correspond to the human movement screen scoring areas, and the

remaining columns may correspond to month long programs specifying a drill in each of the body movement categories. Each of phases 1-6 may be a month long program, and the score 1 and 2 columns may correspond to month long programs of remedial exercises for users having human movement screen scores less than 3 in a particular body movement category. In phase 6, for example, the program includes a renegade row exercise in the core stability movement category, a seated Russian twist exercise in the twist movement category, and so forth. The user may perform the same exercise drills during each exercise session over the month, but the intensity and duration of each exercise drill may change over the month according to a 4 week routine. The computer **102** may also permit the user to swap one drill for an equivalent drill.

(103) If a user receives a human movement screen score of 3 in all categories, the computer **102** may prompt the user to performance exercises shown in the month 1 column. If the user receives a human movement screen score of 1 or 2 in any body movement category, the computer **102** may prompt the user to perform the body movement in the score 1 or score 2 columns for that category. For example, if the user receives a score of 1 in the pull category, the computer **102** may prompt the user to perform the reach roll'n lift exercise in month 1, the Lying T's in month 2, and so forth along that row and the six month program would end at the bent over row exercise from the month 4 column.

(104) In another example, the workout plan may include a baseline workout and six month long programs, examples of which are depicted in FIGS. **21-34**. FIGS. **21-24** describe Workout A, which includes exercises focusing on developing a user's strength and power. FIGS. **25-32** describe Workout B, which includes metabolic focus with exercises to develop a user's speed, agility, and power. FIGS. **33-34** describe Workout C, which includes regeneration exercises such as stretching exercises. Workout A, for example, may be the priority workout if a user only works out once per week, but other workouts may be prioritized as well.

(105) With reference to FIG. **22**, the Workout A program for months 1-2 may include an exercise list, a relationship between workout phases and human movement screen, and phase details. The program may include a dynamic warm-up, followed by exercises in each of the body movement categories. After completed the listed exercises, if the user has a human movement screen score of 14 or more, the program may include a metabolic challenge where the user is prompted to perform an exercise attempting to physically challenge the user (e.g., run as fast as you can, do repetitions until muscle failure, etc.). Thereafter, the program may include regeneration exercises (e.g., stretching).

(106) With reference to FIG. **26**, the workout plan may specify a number of sets, an amount of time of exercise, and an amount of time of rest. For example, in month 1, Workout B specifies 2-3 sets per exercise, where the user works for 20 seconds followed by 20 seconds of rest.

(107) Each month long program over the six month program may be divided into 4 phases each lasting a week, an example of which is depicted in FIG. **35**. Week 1 may be the "introduction week" to introduce the user to an instructional technique and to obtain a baseline physical fitness level. Week 2 may be the "base week" and may involve the user improving their technique as well as increasing workout intensity. Week 3 may be the "overload week" and may involve adding more load (e.g., weight) to a working program and increasing intensity. Week 4 may be the "challenge week" and may involve pushing a user to their maximum. Other 4 week progressions may also be used. Generally, the human body adapts every 4-6 weeks. Some users may adapt at different intervals and the program may be structure to account for these differences. Some programs may also include shorter or longer progressions and the progressions may be determined by analyzing the performance of a user. The computer **102** prompts the user to learn proper technique which leads to progression. Challenges are increased over time by adding repetitions and equipment. As the user conquers a workout, they are given a new routine to learn, providing an intrinsic reward. The computer **102** attempts to learn a user's most effective workout, and reassess a user's performance every 4 weeks.

(108) FIG. 36 illustrates a flow diagram for leading a user through a workout session. The computer **102** may prompt the user to begin a workout session, and ask how long the user has to work out, as seen in FIG. 37. The computer **102** may then provide a summary of the exercise drills the user is going to perform. The computer **102** may provide the how and the why of workouts to help the user get into the mindset of a competitive athlete. Competitive athletes may include professional athletes, college athletes, personal trainers, community leaderboard participants and others. For example, the drills may relate to corrective/core exercises, strength/power exercises, energize/metabolic exercises, regenerate/stretch exercises. The computer **102** may lead the user through a warm up session (as seen in FIG. 38). The warm up session may be a dynamic warm-up designed to warm up the user's muscles in each of the body movement categories. The computer **102** may then provide a demonstration of a first drill (as seen in FIG. 39). The user may then perform the drill in front of the image capturing device **304** as the computer **102** process images of the user. The computer **102** may cause a display to display an image of the user performing the drill (see FIG. 40). The computer **102** may also provide encouragement to the user to keep going (e.g., just 5 more repetitions) as well as feedback. Encouragement may be given after detecting a predetermined number of repetitions (e.g., every 3-5 repetitions), in response to detecting a repetition rate decreases below a level, or other metric.

(109) The feedback may allow the user to compete against their own benchmarks to see improvement in real-time and over time. FIG. 41, for example, illustrates a comparison of a user's form versus desired form. In another example, FIG. 42 illustrates an image of the user exercising with added straight lines to show proper back and hip posture during a deep squat. In a further example, FIGS. 43A-B illustrates graphical user interfaces providing the user with feedback on their form (i.e., Correction needed: straighten knees), and removes the corrective feedback when the user's form improves. Once one drill is completed, the computer **102** may prompt the user to move onto the next drill. FIG. 44, for instance, illustrates a graphical user interface informing the user of the next training category during the workout session.

(110) As shown by the examples in FIG. 45, the system may use data points to determine feedback and motivation to provide to a user during a workout session. The system may also revise a workout session based on the amount of time a user can commit to a workout, as described with reference to FIGS. 46-49. For example, a workout may require 60 minutes, but a user may only have 20 minutes (see FIGS. 46-47). The system may scale down the workout session based on the amount of time a user has (see FIG. 49). Adjustments may include altering an amount of warm up time, a number of sets, a number of repetitions, substitution of exercises, and the like.

(111) The system may also account for missed workout session. If a user is able to make at least one workout session per week, the computer **102** may continue with the scheduled workout program (see, for example, FIG. 20). If the user misses an entire week, the computer **102** may start the user to perform the next workout where they left off. If the user has missed two or more weeks, the computer **102** may have the user repeat the last week they attended. The computer **102** may also send a message to a user's device (e.g., desktop computer, smart phone, laptop computer, etc.) informing of the missed workout, how many workouts have been missed, and how the missed workout has affected reaching the user's goals. The computer **102** may also include in the message a prompt to reschedule the missed workout. In some embodiments users may skip workouts.

(112) FIG. 50 illustrates a flow diagram for providing a user with post-workout information. Upon completion of a workout, the computer **102** may inform the user that the workout session is complete and may cause the trainer avatar to extend a first for a virtual first bump or other congratulatory movement (see FIG. 52). The computer **102** may inform the user of their workout performance and an amount of points associated with the workout (see FIG. 52-53). The computer **102** may also prompt the user to continue exercising (see FIG. 54). The computer **102** may also provide the user with feedback on their form, and may indicate a number of repetitions in the preferred, red, and green zones, discussed above, for each of the exercise drills.

(113) The computer **102** may also calculate a fatigue index indicating how well the user maintained good form over the duration of a drill. For example, the fatigue index may indicate that the user was in the preferred zone for the first 4 repetitions, in the good zone for the next 5 repetitions, and in the red zone for the last repetition.

(114) If the user trains hard during a session, the computer **102** may associate a greater number of points and unlock new workouts. Upon reaching point milestones, the user may unlock workouts and online challenges, or the user may purchase these items online through a gaming console. Other incentives may include obtaining certification as a trainer upon reaching certain fitness milestones. The user may also purchase products from a particular clothing or footwear supplier to increase rewards. For example, a product may have an embedded barcodes or other information that a user may scan or otherwise input to the computer **102** to unlock new training sessions (e.g., a session about stretching for a run). In some embodiments the purchase of certain products may allow a user to unlock new workouts. The new workouts may be related to or use the purchased products.

(115) A display device may present a graphical user interface of a post-workout-dashboard permitting a user to review training data with analysis to view progress and improve future sessions. The user may also elect to post their workout online via social networking (e.g., via a social networking website) or otherwise share their workout sessions. Users may post comments and provide recommendations when reviewing workouts of other users. Users may also post messages to provide motivation to other users. The computer **102** may also post information to a social network when a user improves their fitness level (e.g., Bob improved his fitness level from intermediate to advanced). The computer **102** may have a dynamic recommendation engine that suggests new workouts based on profile and previous training successes. Trainers may also recommend different types of engagements such as joining a challenge or going head to head with a friend. The computer **102** may then suggest a time and date for a next workout session.

(116) FIG. 55 illustrates a flow diagram of a drop-in workout session and FIGS. 56-59 illustrate corresponding graphical user interfaces. A drop-in workout session may be an extra workout session in addition to the ones a user has scheduled for a particular week. A drop-in session may also be where a user opts to do a different workout than the one the scheduled for the user. The computer **102** may adjust future workouts based on the drop-in session. Adjustments may include adding or removing a future workout of a particular type (e.g., Drop-in session is an A workout, so replace a future A workout with a B workout). If a user is avoiding a particular exercise, the computer **102** may identify one or more equivalent exercises and may adjust future workouts to exclude the particular exercise. Other adjustments may also be made.

(117) To initiate a drop-in session, the user may select a drop-in workout session tab of a graphical user interface (see FIG. 56). The computer **102** may ask how long the user has to work out (see FIG. 57) and what type of session the user wants to do, examples of which may include a fully body workout, a strength workout, an endurance workout, a mobility workout, or a balance workout (see FIG. 58). The computer **102** may then perform static postural analysis and/or a body scan. The computer **102** may then provide a summary of the exercise drills the user is going to perform. For example, the drills may relate to corrective/core exercises, strength/power exercises, energize/metabolic exercises, regenerate/stretch exercises. The computer **102** may lead the user through a warm up session, and then proceed to a first drill (see FIG. 59). A display device may present a demonstration of the drill. The user may then perform the drill in front of the image capturing device as the computer **102** process images of the user. The computer **102** may also provide encouragement to the user to keep going (e.g., just 5 more repetitions). The computer **102** then presents feedback to the user. Once one drill is completed, the computer **102** may prompt the user to move onto the next drill. After completion of the last drill, the computer **102** may then update a user's points and provide them with a post-workout summary.

(118) A challenge session may be where a user competes against a ghost of their previous workout

or another user. For example, the computer **102** may store video of a user performing a set of exercises, as well as performance metrics. The display may present the video of the user where the user appears translucent, and hence is denoted as a ghost. The display may overlay video recorded by the image capturing device for comparison with the ghost. The computer **102** may provide a demonstration of the challenge, and the user may perform the challenge. Upon completion of the challenge, the computer **102** may display the challenge results.

(119) The user may also create their own challenges and workout sessions for more focused training or sharing with a social network. The user may receive points, money, or other incentives based on a number of other users who download a user created workout session. The user may also cause the computer **102** to request ghost workouts from friends or pros to aid or compare.

(120) Challenges may also be against multiple players at a single location (e.g., house), or via a network. FIG. **60** illustrates an exemplary challenge drill. The computer **102** may identify other users having similar fitness levels or users may challenge other users. Multiple players may simultaneously participate in a workout session at the same location, or may sequentially participate where a ghost of a player who completed a session is displayed onscreen competing against a later player. Also, the computer **102** may join an online challenge where a user may compete against another player who is also online. The online challenge may permit competitions with multiple other players. Competitions may be organized by age group, fitness level, invitation only, achieving a certain point levels, or in other manners. A ghost of the leader may be presented by the displays of all other challengers. The computer **102** may also cause a display to present a leader board showing how a user compares to other participants. Performance data, such as personal bests, may be communicated graphically and through audio devices.

CONCLUSION

(121) Aspects of the embodiments have been described in terms of illustrative embodiments thereof. Numerous other embodiments, modifications and variations within the scope and spirit of the appended claims will occur to persons of ordinary skill in the art from a review of this disclosure. For example, one of ordinary skill in the art will appreciate that the steps illustrated in the illustrative figures may be performed in other than the recited order, and that one or more steps illustrated may be optional in accordance with aspects of the embodiments.

Claims

1. A computer implemented method comprising: capturing, with a wrist-worn sensor, motion parameter data of a user performing an athletic movement; identifying a form of the user based on an identified sensor region and based on tracking motion parameter data in the identified sensor region; generating, at a processor, one or more human movement screen scores based on the form of the user and based on the at least a first portion of the motion parameter data of the user performing the athletic movement; providing a feedback display that includes a graphical comparison of the form of the user versus a desired form and an identification of the one or more human movement screen scores; generating, by the processor, a personalized exercise program based on the one or more human movement screen scores, and comprising a plurality of workouts; in response to determining that a current workout session is not a first workout session for the user, determining a previous human movement screen score from a previous workout, a duration since the previous workout, and a number of missed workouts during the duration; selecting, by the processor, from the plurality of workouts in the personalized exercise program and based the duration since the previous workout and the number of missed workouts during the duration, a current workout, wherein the current workout includes one or more exercises selected based on the previous human movement screen score; determining a length of time that the user has available for completing the current workout of the plurality of workouts in the personalized exercise program; and in response to determining that a length of time associated with the current workout

is greater than a length of time the user has available, adjusting the current workout.

2. The computer implemented method of claim 1, wherein the capturing motion parameter data further comprises capturing, with an image capture device, images of the user performing the athletic movement.

3. The computer implemented method of claim 2, wherein the image capture device comprises a plurality of cameras.

4. The computer implemented method of claim 1, wherein the generating one or more human movement screen scores comprises generating, at a processor, a human movement screen score based on the wrist-worn sensor and a second sensor.

5. The computer implemented method of claim 1, wherein the wrist-worn sensor, comprises an accelerometer.

6. The computer implemented method of claim 1, wherein the capturing motion parameter data further comprises capturing motion parameters with a footwear-worn accelerometer.

7. The computer implemented method of claim 1, wherein the wrist-worn sensor comprises an orientation determining sensor.

8. The computer implemented method of claim 1, wherein generating the personalized exercise program is based on the one or more human movement screen scores and an input of the user.

9. The computer implemented method of claim 8, wherein the input of the user comprises a time commitment.

10. The computer implemented method of claim 8, wherein the input of the user comprises a number of exercise sessions in a predetermined time period.

11. The computer implemented method of claim 1, wherein the generating one or more human movement screen scores comprises evaluating a form of the user by identifying locations of body parts of the user at different times.

12. The computer implemented method of claim 1, wherein the wrist-worn sensor comprises a display.

13. The computer implemented method of claim 1, wherein the personalized exercise program includes a ghost workout received from a friend of the user.

14. The computer implemented method of claim 1, wherein adjusting the current workout comprises substitution of one or more exercises within the current workout.

15. The computer implemented method of claim 1, wherein adjusting the current workout comprises altering one of: an amount of warmup time, a number of sets of an exercise of the current workout, or a number of repetitions of an exercise of the current workout.

16. The computer implemented method of claim 1, wherein the feedback display further comprises a highlighting of a body part of the user associated with improper form and a correction to a form of the user.

17. An apparatus configured to be worn on an appendage of a user, comprising: at least one processor; a display device; a sensor device; a non-transitory computer-readable medium comprising computer-executable instructions that when executed by the at least one processor cause the at least one processor to: capture, from the sensor device, motion parameter data of the user performing an athletic movement; identify a form of the user based on an identified sensor region and based on tracking motion parameter data in the identified sensor region; generate, by the at least one processor, one or more human movement screen scores based on the form of the user and based on the at least a first portion of the motion parameter data of the user performing the athletic movement; provide a feedback display that includes a graphical comparison of the form of the user versus a desired form and an identification of the one or more human movement screen scores; generate, by the at least one processor, a personalized exercise program based on the one or more human movement screen scores, and comprising a plurality of workouts; in response to determining that a current workout session is not a first workout session for the user, determine a previous human movement screen score from a previous workout, a duration since the previous

workout, and a number of missed workouts during the duration; determine by the processor from the plurality of workouts in the personalized exercise program and based on both the duration since the previous workout and the number of missed workouts during the duration, a current workout, wherein the current workout includes one or more exercises selected based on the previous human movement screen score; determine whether a length of time associated with the current workout is greater than a length of time the user has available for completing the current workout; and automatically adjust the current workout based on determining that the length of time associated with the current workout is greater than a length of time the user has available for completing the current workout, wherein adjusting the current workout comprises altering one or more of: an amount of warmup time, a number of sets of an exercise of the current workout, and a number of repetitions of an exercise of the current workout.

18. The apparatus of claim 17, wherein the capturing comprises obtaining accelerometer data from an accelerometer located on the apparatus.

19. The apparatus of claim 18, wherein the accelerometer data comprises data from a plurality of axes.

20. The apparatus of claim 18, wherein the computer-readable medium further comprises computer-executable instructions that when executed by the at least one processor cause the at least one processor to: collect motion parameters of the user from a footwear-worn accelerometer; and wherein generating the one or more human movement screen scores is based upon at least data obtained from the footwear-worn accelerometer and the motion parameters captured from the sensor device.

21. The apparatus of claim 18, wherein the sensor device comprises a gyroscope.

22. The apparatus of claim 17, wherein the personalized exercise program includes a ghost workout received from a friend of the user.

23. A non-transitory computer-readable medium comprising computer-executable instructions that when executed by a processor, cause the processor to perform at least: capturing, with a body-worn sensor, motion parameter data of a user performing an athletic movement; identifying a form of the user based on an identified sensor region and based on tracking motion parameter data in the identified sensor region; generating, with at least one processor, one or more human movement screen scores based on the form of the user and based on the at least a first portion of motion parameter data of the user performing the athletic movement; providing a feedback display that includes a graphical comparison of the form of the user versus a desired form and an identification of the one or more human movement screen scores; generating, at the at least one processor, a personalized exercise program based on the one or more human movement screen scores, and comprising a plurality of workouts; in response to determining that a current workout session is not a first workout session for the user, determining a previous human movement screen score from a previous workout, a duration since the previous workout, and a number of missed workouts during the duration, wherein the current workout session includes one or more exercises selected based on the previous human movement screen score; determining, by the at least one processor, a next workout from the plurality of workouts based on the previous human movement screen score from the previous workout, the duration since the previous workout, and the number of missed workouts during the duration; and in response to determining that a length of time associated with the next workout is greater than a length of time the user has available for completing the next workout, adjusting the next workout.
